ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Diversity in the Living World** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** biodiversity	Biodiversity means diversity of life.
2	**(b)** parallel venation	Monocot leaves have parallel veins.
3	**(b)** herbivores	Cows are herbivores.
4	**(c)** habitat	A habitat provides food and shelter.
5	**(c)** tap root system	Mustard and carrot have tap roots.
6	**(d)** carnivores	Lions are carnivores.
7	**(c)** herbs	Herbs have tender stems.
8	**(b)** fibrous root system	Many similar thin roots arise from the base.
9	**(d)** omnivores	Humans and crows are omnivores.
10	**(c)** shrubs	Rose and hibiscus are shrubs.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The variety of living organisms on Earth (1); It includes plants, animals and microorganisms (1).	1+1
12	Tap root: carrot, mustard (1); Fibrous: wheat, grass (1).	1+1
13	Long legs and padded feet for sand (1); It can go long without water (1).	1+1
14	Herbs have soft green stems and are small (1); Shrubs have hard woody stems branching near the base (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Tomato: herb (1); Rose: shrub (1); Banyan: tree (1).	1+1+1
16	Goat: herbivore (1); Tiger: carnivore (1); Bear: omnivore (1).	1+1+1
17	They make food by photosynthesis (1); Leaves capture sunlight (1); Animals depend on this food (1).	1+1+1
	OR: Tap root has one main root with branches (1); Fibrous roots are many thin roots of similar size (1); Examples: carrot and wheat (1).	1+1+1
18	Herb: spinach (1); Shrub: hibiscus (1); Tree: neem (1).	1+1+1
	OR: Supports leaves, flowers and fruits (1); Carries water and minerals (1); Carries food to different parts (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Herbs: soft green stem, e.g. spinach (1); Shrubs: hard branching near the base, e.g. hibiscus (1); Trees: thick trunk, e.g. mango (1); Climbers: need support, e.g. pea (1); Creepers: spread on ground, e.g. pumpkin (1).	1+1+1+1+1
	OR: Blade or lamina is the broad part (1); Petiole is the stalk (1); Midrib is the main vein (1); Veins carry water and food (1); Pattern of veins is venation (1).	1+1+1+1+1
20	Tap root: one main root, e.g. carrot (1); Fibrous: many similar roots, e.g. wheat (1); Reticulate venation: net-like, e.g. peepal (1); Parallel venation: veins parallel, e.g. grass (1); Roots of grass are fibrous, matching parallel veins (1).	1+1+1+1+1
	OR: Collect fallen leaves of different plants (1); Press them in a book (1); Paste them on paper (1); Note names and venation (1); Group similar leaves together (1).	1+1+1+1+1

Mistakes to avoid:

- Calling any small plant a herb; a shrub is also small but has a **hard stem** branching near the base.
- Mixing up creepers and climbers: creepers spread on the ground, climbers go up a support.
- Writing that tap roots go with parallel venation; tap root usually goes with **reticulate** venation, fibrous with parallel.
- Confusing **habitat** (the place) with **adaptation** (the feature).
- Saying a camel stores water in its hump; the hump stores fat.
- Thinking venation means the roots; it is the pattern of veins in the **leaf**.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Diversity in the Living World** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** leaf	Leaves carry out photosynthesis.
2	**(b)** crow	Crows eat grains and flesh.
3	**(d)** trees	Mango and neem are trees.
4	**(c)** root	Roots anchor and absorb.
5	**(a)** lion	Lions hunt other animals.
6	**(c)** climbers	Climbers use tendrils or other parts to climb.
7	**(b)** flower	Flowers lead to fruits.
8	**(b)** fly	They reduce weight and aid flight.
9	**(c)** creepers	Pumpkin and watermelon creep.
10	**(c)** can live long without water	It stores fat and conserves water.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It reduces water resistance (1); It helps them swim fast (1).	1+1
12	Crow (1); Human or bear (1).	1+1
13	The place where an organism lives (1); Example: a pond is the habitat of fish (1).	1+1
14	Climbers climb with support (1); Creepers spread on the ground (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Parallel: veins run side by side (1); Reticulate: veins form a net (1); Examples: grass; mango (1).	1+1+1
16	It is a special feature that helps survival (1); Camel in desert (1); Fish gills in water (1).	1+1+1
17	It is easier to study them (1); It shows similarities and differences (1); It helps us to identify new ones (1).	1+1+1
	OR: Streamlined body (1); Gills for breathing (1); Fins for movement (1).	1+1+1
18	Terrestrial: on land (1); Aquatic: in water (1); Examples: forest; pond (1).	1+1+1
	OR: Root: absorbs water (1); Stem: supports (1); Leaf: makes food (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Camel stores fat in the hump (1); Camel has broad feet (1); Fish has gills and fins (1); Polar bear has thick fur (1); Polar bear has a fat layer (1).	1+1+1+1+1
	OR: Webbed feet in frog and duck (1); Lungs and moist skin in frog (1); Crocodile has strong tail for swimming (1); Nostrils placed high on head (1); Legs for land movement (1).	1+1+1+1+1
20	Herbivores: cow, deer (1); Carnivores: lion, tiger (1); Omnivores: crow, bear (1); Based on food habits (1); Teeth differ accordingly (1).	1+1+1+1+1
	OR: Bird: wings, feathers, hollow bones (1); Fish: fins (1); Fish: streamlined body (1); Snake: long flexible body (1); Snake moves by muscles of body and scales (1).	1+1+1+1+1

Mistakes to avoid:

- Calling any small plant a herb; a shrub is also small but has a **hard stem** branching near the base.
- Mixing up creepers and climbers: creepers spread on the ground, climbers go up a support.
- Writing that tap roots go with parallel venation; tap root usually goes with **reticulate** venation, fibrous with parallel.
- Confusing **habitat** (the place) with **adaptation** (the feature).
- Saying a camel stores water in its hump; the hump stores fat.
- Thinking venation means the roots; it is the pattern of veins in the **leaf**.

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Diversity in the Living World** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** aquatic animals	Aquatic means water-living.
2	**(b)** pumpkin	Pumpkin spreads along the ground.
3	**(d)** frog	Frogs are amphibians.
4	**(c)** terrestrial animals	Terrestrial means land-dwelling.
5	**(a)** venation	Venation is the arrangement of veins.
6	**(a)** gills	Gills take oxygen from water.
7	**(b)** adaptation	Adaptation helps survival.
8	**(b)** a net-like pattern	Mango and peepal leaves show it.
9	**(b)** move easily through water	It reduces water resistance.
10	**(c)** stay warm in the cold	Fur insulates from cold.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	To make the body light (1); It helps them to fly (1).	1+1
12	Arrangement of veins in a leaf (1); Parallel and reticulate (1).	1+1
13	Herbivores eat only plants (1); Carnivores eat only animals (1).	1+1
14	Water differs from land in oxygen and movement (1); Gills and fins help them survive (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Feathers and wings (1); Hollow light bones (1); Streamlined body (1).	1+1+1
16	It insulates from cold (1); Fat stores energy (1); It lives in Arctic ice and snow (1).	1+1+1
17	Animals that fly in air (1); Feathers or wings (1); Birds and bats (1).	1+1+1
	OR: Store fat in hump (1); Long eyelashes keep out sand (1); Wide padded feet help walking on sand (1).	1+1+1
18	Have lungs and moist skin for breathing (1); Webbed feet for swimming (1); Strong legs for jumping on land (1).	1+1+1
	OR: Shrub is shorter (1); Tree has a thick trunk (1); Shrub branches near the base (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Variety of living organisms (1); It keeps nature in balance (1); It gives food and medicines (1); Protect forests (1); Avoid hunting and pollution (1).	1+1+1+1+1
	OR: Cactus has thick stem for water (1); Cactus has spines instead of leaves (1); Lotus has long stalks and floating leaves (1); Pine has cone shape that sheds snow (1); Pine has needle-like leaves (1).	1+1+1+1+1
20	Terrestrial: legs for walking (1); Aquatic: fins and gills (1); Aerial: wings and feathers (1); Examples: dog, fish, parrot (1); Body shapes suit the habitat (1).	1+1+1+1+1
	OR: Food chains depend on many species (1); Many plants give medicines (1); Forests clean the air (1); Loss of one species affects others (1); Future generations need these resources (1).	1+1+1+1+1

Mistakes to avoid:

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- Mixing up creepers and climbers: creepers spread on the ground, climbers go up a support.
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